Worksheet — Air Resistance Simulation

Use with *Simulation - Air resistance.html* · A.1 Kinematics: qualitative effect of air resistance on a projectile (drag $F_d = \frac{1}{2}\rho C_d A v^2$); goes with throwing / filming a ping-pong ball or shuttlecock beside a tennis ball

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/15-air-resistance** in a browser (offline: *Simulation - Air resistance.html*). Choose an object, the speed and the angle, then ▶️ **Launch**. The **solid** path has air resistance; the **dashed** path is the same launch in a vacuum.

Part 1 before the lab (≈ 15 min) · **Part 2** during the lab (≈ 15 min) · **Part 3** homework (≈ 30 min).

Drag always points opposite to the velocity and grows with v^2. At SL you must **describe** its effects; the numbers help you see them.

Scan to open

Part 1 — Pre-lab: predict, then test

Tennis ball, 25 m s^{-1} at 45°. Predict **before** you launch, then test and explain.

Question	My prediction (before)	What the simulation showed	Physics explanation
1. How does the real path differ from the vacuum path? Sketch both.			
2. On the way up, is the acceleration smaller than, equal to or larger than g?			
3. Which takes longer: going up or coming down?			
4. In which direction does the drag act at the top of the path?			

5. Calculate the range in a vacuum for 25 m s^{-1} at 45°. By what percentage is the real range shorter?

Part 2 — During the lab: light and heavy balls

Class. Throw a ping-pong ball (or shuttlecock) and a tennis ball at about the same speed and angle; film them if you can. **Simulation:** 10 m s^{-1} at 45° for both.

	ping-pong ball	tennis ball	vacuum
range R / m			
max height H / m			
landing angle (steep / similar)			
terminal speed v_t / m s^{-1}			—

1. Describe the shape of the ping-pong ball's path. How is it different from the tennis ball's?

2. Explain why the ping-pong ball is affected much more. Use F_d/m.

3. Sketch the path of a shuttlecock from the class (or video) and compare with the simulation.

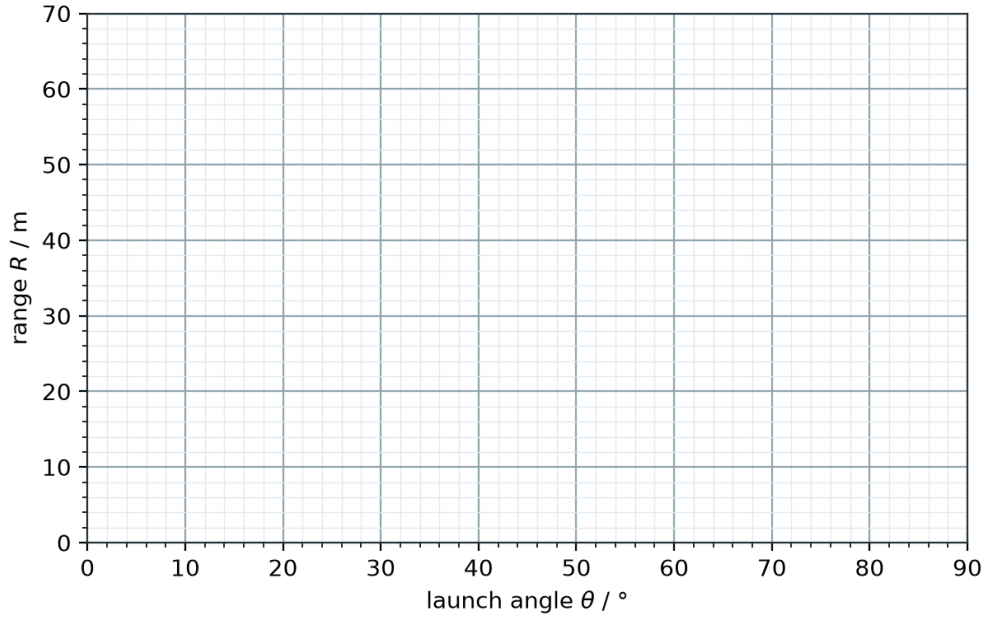
Sketch

Part 3 — Homework: analyse

Tennis ball, 25 m s^{-1}. Record each angle and copy both ranges.

θ / °	20	30	35	40	45	50	60
R with air / m							
R vacuum / m							

1. Plot both ranges against θ on the same axes.



2. At what angle is the range with air largest? Why is it less than 45°?

3. Use $mg = \frac{1}{2}\rho C_d A v_t^2$ to calculate the terminal speed of a ping-pong ball (m = 2.7 g, d = 40 mm, C_d = 0.47, ρ = 1.20 kg m^{-3}).

4. Calculate the drag force on a tennis ball (m = 58 g, d = 67 mm, C_d = 0.55) at 25 m s^{-1} and compare it with its weight.

5. For which object is the vacuum model best: ping-pong ball, shuttlecock, tennis ball or steel ball? Explain.